Peer Feedback

- For the peer feedback we paired up with Bryce Bosworth and Kristin Knippenberg. Their project investigates the association between suicide-related ER visits and air pollution.
- Team members present:
    - Mason Buckman
    - Kim Lanaghen
    - Lindsey McGuire
    - Clara Garcia
- Feedback received:
    - Take into consideration the impact of the Bear River as a major feeder to the Great Salt Lake, as its flow likely plays a significant role in lake level changes.
    - Consider incorporating precipitation data from additional weather stations across Northern Utah, beyond the primary region of interest.
    - Look into other variables that drive water consumption and influence lake levels, such as agricultural use (e.g., alfalfa farming).
    - What does a given level of rainfall predict about how much the lake grew or shrank? What other variables contributed to changes that aren't captured linearly?
    - To narrow down the methodology, start by selecting a few key variables to focus on, then let the data guide which methods to use and what further questions to ask.
    - Consider comparing the Great Salt Lake to similar lakes that have experienced drought or shrinkage, such as those in the Middle East or California, to help validate findings.
    - Compare model to/replicate other studies that have been done. Then, show the difference between models.
        - With the data that we have, can we get similar data to the model that was already done?
        - Why was a particular model chosen for prior studies? How are we looking at things differently?
    - What does that rainfall mean for everything else? Flooding/ice age, when in history have the conditions be closest to those types of events? Consider sampling those conditions into the model.
    - Be mindful of spurious correlations.
- How do we plan to address feedback?
    - Some of the feedback focused on selecting variables that contribute to lake levels. Our original plan was to focus primarily on snowpack and precipitation data. After further research, we have

decided to expand our variable selection to also include inflow tributary data from three major sources: the Bear River, Weber River, and Jordan River. In addition to Great Salt Lake elevation data, we will construct our own evaporation dataset using inflow tributary, precipitation, and lake elevation data, calculated as follows:

- $E_t = V_{(t-1)} + R_t + P_t - V_t$
    - V_t = current month's lake volume
    - $V_{(t-1)}$ = prior month's lake volume
    - R_t = that month's total inflow
    - P_t = that month's precipitation on the lake surface
    - E_t = that month's evaporation
- We will compare our evaporation data against findings from Bigalke et al. (2025), who examined the contributing variables to the 2022 record low Great Salt Lake volume:
    - Bigalke, S., Loikith, P., & Siler, N. (2025). Explaining the 2022 record low Great Salt Lake volume. *Geophysical Research Letters, 52*, e2024GL112154. https://doi.org/10.1029/2024GL112154

- In response to feedback about replicating an existing model, we will apply the K-Nearest Neighbor (K-NN) bootstrap method to predict future lake levels, following the approach outlined in Lall & Sharma (1996):
    - Lall, U., & Sharma, A. (1996). A nearest neighbor bootstrap for resampling hydrologic time series. *Water Resources Research, 32*(3), 679–693. https://doi.org/10.1029/95WR02966